



National Forensic Sciences University, Gandhinagar

School of Cyber Security and Digital Forensics

Assignment: Practical - 5 : Wireshark

Name: Aastha Thakker

Roll No. 250103002004

Sem: 2

Subject: Network Security

Course: MSc Cyber Security

Batch: 2025 -2027

Evidence_01.pcap

After being released on bail, Ann Dercover disappears! Fortunately, investigators were careful monitoring her network activity before she skipped town. “We believe Ann may have communicated with her secret lover, Mr. X, before she left,” says the police chief. “The packet capture may contain clues to her whereabouts.” You are the forensic investigator. Your mission is to figure out what Ann emailed, where she went, and recover evidence, including:

1. What is Ann’s email address? Ans: sneakyg33k@aol.com

Wireshark packet capture showing an SMTP session. The selected packet (66) is an SMTP AUTH LOGIN command. The payload shows the email address 'sneakyg33k@aol.com'.

2. What is Ann’s email password? Ans: 558r00lz

Wireshark packet capture showing an SMTP session. The selected packet (66) is an SMTP AUTH LOGIN command. The payload shows the email address 'sneakyg33k@aol.com'.

Base64 decode tool interface showing the decoded password: 558r00lz.

6. **What is the MD5sum of the attachment Ann sent to her secret lover? Ans:**
d3e4673749343d1e0fba06552bc0b035
7. **In what CITY and COUNTRY is their rendezvous point? Ans:** City:Playa del Carmen
Country: Mexico

Evidence_02.pcap

You are investigating suspicious activity on a Windows machine suspected to be infected with malware. A PCAP file containing network traffic was captured during the suspected activity. The captured traffic reveals that the malware attempted to exfiltrate sensitive information through email.

1. Identify the first SMTP packet associated with the malicious email transmission and provide the date and time in UTC. Ans: 2023/05/01 22:51:29.57 [SYN] packet transmission was done at this time as it initiates the conversation between client and server

No.	Time	Source	Destination	Protocol	Length	Info
646	30.412813	204.11.58.28	192.168.1.27	SMTP	232	S: 220-bh-41.webhostbox.net ESMTPEXIM 4.95 #2 Thu, 05 Jan 2023 22:51:30 +0000 We do not authorize the use of
647	30.413109	192.168.1.27	204.11.58.28	SMTP	76	C: EHLO DESKTOP-WIN11PC
649	30.512841	204.11.58.28	192.168.1.27	SMTP	231	S: 250-bh-41.webhostbox.net Hello DESKTOP-WIN11PC [173.66.46.112] SIZE 52428800 8BITIME PIPELINING PIPE
650	30.514513	192.168.1.27	204.11.58.28	SMTP	99	C: AUTH login User: bWfYa2V0aW5nQHRyYW5zZ2VhcnI5bGw=
652	30.620821	204.11.58.28	192.168.1.27	SMTP	72	S: 334 UGfzc3dvcM06
653	30.622428	192.168.1.27	204.11.58.28	SMTP	72	C: Pass: TUB2c3cwcmQjNjIx
655	30.805037	204.11.58.28	192.168.1.27	SMTP	84	S: 235 Authentication succeeded
656	30.805955	192.168.1.27	204.11.58.28	SMTP	90	C: MAIL FROM:<marketing@transgear.in>
658	30.909911	204.11.58.28	192.168.1.27	SMTP	62	S: 250 OK
659	30.910382	192.168.1.27	204.11.58.28	SMTP	94	C: RCPT TO:<zaritkt@arhitektondizajn.com>
661	31.044757	204.11.58.28	192.168.1.27	SMTP	68	S: 250 Accepted
662	31.045187	192.168.1.27	204.11.58.28	SMTP	60	C: DATA
664	31.150936	204.11.58.28	192.168.1.27	SMTP	110	S: 354 Enter message, ending with "." on a line by itself
665	31.153380	192.168.1.27	204.11.58.28	SMTP	303	C: DATA fragment, 249 bytes
667	31.153587	192.168.1.27	204.11.58.28	SMTP	1077	C: DATA fragment, 1023 bytes
669	31.153750	192.168.1.27	204.11.58.28	SMTP	669	C: DATA fragment, 615 bytes
671	31.153954	192.168.1.27	204.11.58.28	SMTP	56	C: DATA fragment, 2 bytes
675	31.364006	204.11.58.28	192.168.1.27	SMTP	82	S: 250 OK Id=IpDZ59-000wFk-B4
680	129.075407	192.168.1.27	204.11.58.28	SMTP	60	C: QUIT
688	129.768059	204.11.58.28	192.168.1.27	SMTP	99	S: 221 bh-41.webhostbox.net closing connection

```

Time: 01/05/2023 22:51:26<br>User Name: windows11user<br>Computer=
Name: DESKTOP-WIN11PC<br>OSFullName: Microsoft Windows 11 Pro<br>
>CPU: Intel(R) Core(TM) i5-13600K CPU @ 5.10GHz<br>RAM: 32165.83 =
MB<br>IP Address: 173.66.46.112<br><hr>URL:imap://mail.windows11u=
sers.com<br>=0D=0AUsername:admin@windows11users.com<br>=0D=0APass=
word:EBj%U7-p@q4NW<br>=0D=0AApplication:Thunderbird<br>=0D=0A<br>=
=0D=0AURL:smtp://mail.windows11users.com<br>=0D=0AUsername:admin@=
windows11users.com<br>=0D=0APassword:EBj%U7-p@q4NW<br>=0D=0AApplicat=
ion:Thunderbird<br>=0D=0A<br>=0D=0AURL:webmail.windows11users.com=

```

Time	Source	Destination	Protocol	Length	Info
643 2023/05/01 22:51:29.574359	192.168.1.27	204.11.58.28	TCP	60	51958 → 587 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
644 2023/05/01 22:51:29.675200	204.11.58.28	192.168.1.27	TCP	58	587 → 51958 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
645 2023/05/01 22:51:29.675531	192.168.1.27	204.11.58.28	TCP	54	51958 → 587 [ACK] Seq=1 Ack=1 Win=64240 Len=0
646 2023/05/01 22:51:30.494183	204.11.58.28	192.168.1.27	SMTP	232	S: 220-bh-41.webhostbox.net ESMTPEXIM 4.95 #2 Thu, 05 Jan 2023 22:51:30
647 2023/05/01 22:51:30.494559	192.168.1.27	204.11.58.28	SMTP	76	C: EHLO DESKTOP-WIN11PC
648 2023/05/01 22:51:30.494697	204.11.58.28	192.168.1.27	TCP	54	587 → 51958 [ACK] Seq=179 Ack=23 Win=64240 Len=0
649 2023/05/01 22:51:30.594211	204.11.58.28	192.168.1.27	SMTP	231	S: 250-bh-41.webhostbox.net Hello DESKTOP-WIN11PC [173.66.46.112] SIZ
650 2023/05/01 22:51:30.595883	192.168.1.27	204.11.58.28	SMTP	99	C: AUTH login User: bWfYa2V0aW5nQHRyYW5zZ2VhcnI5bGw=

2. At which OSI layer does the malicious SMTP communication begin? Ans: Application Layer
3. Determine the identity of the infected host, including:
 - (a) Internal IP address: 192.168.1.27 (Private IP, which was infected)

(b) **MAC address:** bc:ea:fa:22:74:fb

(c) **The host or computer name of the infected Windows:** DESKTOP-WIN11PC

(d) **System's RAM size:** 32 MB

(e) **Processor specifications:** Inter(R) Core(™) i5-13600k CPU @ 5.10GHz

Time: 01/05/2023 22:51:26

User Name: windows11user

Computer Name: DESKTOP-WIN11PC

OSFullName: Microsoft Windows 11 Pro

CPU: Intel(R) Core(TM) i5-13600K CPU @ 5.10GHz

RAM: 32165.83 MB

IP Address: 173.66.46.112

4. **Describe the types of sensitive data that were exfiltrated. Ans:** Various usernames and passwords of sites like linkedin, amazon, coco-cola etc have been seen.

URL: <https://www.linkedin.com/>

Username: admin@windows11users.com

Password: TqQPvG#0g%\$ga_q51

Application: Edge Chromium

URL: <https://www.amazon.com/ap/signin>

Username: admin@windows11users.com

Password: 3Fo76#PTf4P\$Im!9mkLso69eT

Application: Edge Chromium

URL: <https://www.target.com/login>

Username: windows11user

Password: c\$Kl3wO!e#i7A&!L2

Application: Edge Chromium

URL:smtp://mail.windows11users.com
Username:admin@windows11users.com
Password:EBj%U7-p@q4NW
Application:Thunderbird

URL:webmail.windows11users.com
Username:admin@windows11users.com
Password:EBj%U7-p@q4NW
Application:Edge Chromium

URL:https://login.us.coca-cola.com/
Username:admin@windows11users.com
Password:Zp61-7\$r#J_iLpCYV&jKr
Application:Edge Chromium

URL:imap://mail.windows11users.com
Username:admin@windows11users.com
Password:EBj%U7-p@q4NW
Application:Thunderbird

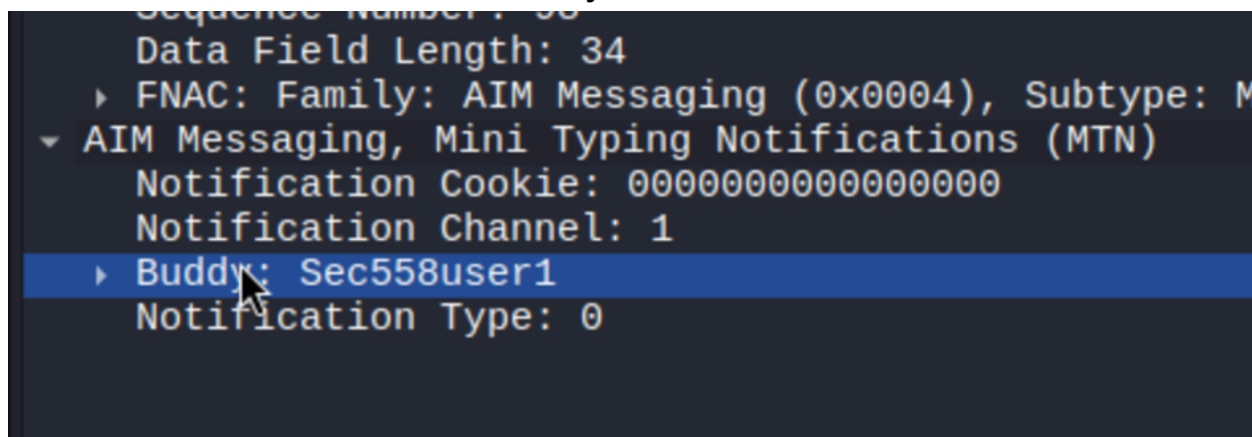
URL:smtp://mail.windows11users.com
Username:admin@windows11users.com
Password:EBj%U7-p@q4NW
Application:Thunderbird

URL:https://myaccount.nytimes.com/auth/login
Username:admin@windows11users.com
Password:u*N21Or650yBsp45awSa
Application:Edge Chromium

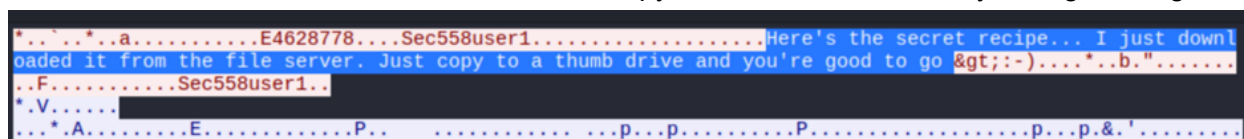
Evidence_03.pcap

Anarchy-R-Us, Inc. suspects that one of their employees, Ann Dercover, is really a secret agent working for their competitor. Ann has access to the company's prized asset, the secret recipe. Security staff are worried that Ann may try to leak the company's secret recipe. Security staff have been monitoring Ann's activity for some time, but haven't found anything suspicious—until now. Today, an unexpected laptop briefly appeared on the company's wireless network. Staff hypothesize it may have been someone in the parking lot, because no strangers were seen in the building. Ann's computer (192.168.1.158) sent IMs over the wireless network to this computer. The rogue laptop disappeared shortly thereafter. "We have a packet capture of the activity", said security staff, "but we can't figure out what's going on. Can you help?" You are the forensic investigator. Your mission is to figure out who Ann was IM-ing, what she sent, and recover evidence, including:

1. What is the name of Ann's IM buddy? Ans: Sec55user1



2. What was the first comment in the captured IM conversation? Ans: I just downloaded it from the file server. Just copy it to a thumb drive and you're good to go...



3. What is the name of the file Ann transferred? Ans: recipe.docx



4. What is the magic number of the file you want to extract (first four bytes)? Ans: PK -ascii (50 4b 03 04)

recipe.docx	-Untitled-	recipe.docx	
50 4B 03 04 3D 7F 01 00	00 A4 05 00 00 13 00 08	PK..=Δ...ñ.....	
02 5B 43 6F 6E 74 65 6E	74 5F 54 79 70 65 73 5D	.[Content_Types]	
2E 78 6D 6C 20 A2 04 02	28 A0 00 02 00 00 00 00	.xml ó..(á.....	
00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00		
00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00		
00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00		

5. What was the MD5sum of the file? Ans: 8350582774e1d4dbe1d61d64c89e0ea1

MD5 (recipe.docx) = 8350582774e1d4dbe1d61d64c89e0ea1

6. What is the secret recipe? Ans: Disaster

Recipe for Disaster:

1 serving

Ingredients:

4 cups sugar

2 cups water

In a medium saucepan, bring the water to a boil. Add sugar. Stir gently over low heat until sugar is fully dissolved. Remove the saucepan from heat. Allow to cool completely. Pour into gas tank. Repeat as necessary.